MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 3XP1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Light output	Rg	Zd	Alt
Unit:	V	Vdc	v	Vdc	Vdc	ftL	Meg	Meg	ft
Maximum:	6.9	0,-200	550	1,100	2,750	---	1.5	1.0	10,000
Minimum:	5.7	---	---	---	1,000 (see note 1)	20 (see note 2)	---	---	---
TEST CONDITIONS:	6.3	Adj	---	Focus	2,000	---	---	---	---

GENERAL:

Qualification - Required

Ⓓ Denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
1141	Pressure (implosion)	-		---	---	---	---
5111	Vibration	-		Width	---	1.0	mm
(D) 1331	Direct-interelectrode capacitance	-	g1 to all	Cg1	---	7	pF
			D1 to D2	C1D2	---	4	pF
			D3 to D4	C3D4	---	4	pF
			D1 to all except D2	CD1	---	4	pF
			D2 to all except D1	CD2	---	4	pF
			D3 to all except D4	CD3	---	4	pF
			D4 to all except D3	CD4	---	4	pF
(D) 5101	Neck and bulb alinement (electrostatic types)	-		Dia	---	1.5	Inch
(D) 5246	Focusing voltage (zero-bias)	-	Ec1 = 0	Eb1	400	---	Vdc
(D) 5248	Deflection-factor uniformity	-		---	---	---	---
(D) 1216	Base material insulating quality	-		---	---	---	---
	Quality conformance inspection, part 1						
5201	Electrode current (cathode)	-	Light = 20 ftL	Ik	---	1,000	uAdc
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross"	3	Light = 20 ftL	---	---	---	---
5101	Base alinement (electrostatic types)	-	+3D4; pin No. 5	---	---	---	---
---	Angle, bulb, and trace	-	+1D2; bulb wall	---	---	1.5	Degrees
5106	Screen and faceplate blemishes	-		---	---	---	---
5221	Light output	2,3		Light	20	---	ftL
5223	Modulation	-	Light = 20 ftL	ΔEc1	---	38	Vdc
5231	Spot position (electro-static deflection)	-		---	---	15.0	mm
5231	Spot displacement (leakage)	-		Displ	---	7.0	mm
5241	Grid cutoff voltage	-		Ec1	---	-67.5	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	-		---	---	---	---
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
5201	Electrode current (anode No. 1)	-	Ec1 = 0	Ib1	-15	10	uAdc

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued						
5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
5101	Angle between traces	-		---	---	---	---
5216	Cathode illumination	-		---	---	---	---
5216	Stray light emission (conventional types)	-	Eb2 = 2,750 Vdc	---	---	---	---
5226	Line width A (electro-static deflection)	-	Light = 20 ftL	Width	---	0.65	mm
5226	Line width B (electro-static deflection)	-	Light = 20 ftL	Width	---	0.65	mm
5246	Focusing voltage at cutoff	-	Ec1 = cutoff	Eb1	---	700	Vdc
5248	Deflection factor	-	1D2	DF	68	92	Vdc/in.
5248	Deflection factor	-	3D4	DF	28	38	Vdc/in.
1111	Base pin solder depth	-		---	---	---	---
1101	Secureness of base, cap, or insert	-		---	---	---	---
(D) 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group C; light = 20 ftL; Eb2 = 2,750 Vdc; t = 500 hrs (min)	---	---	---	---
---	Life-test end points:						
5226	Line width A	-	Light = 16 ftL	Width	---	0.65	mm
5226	Line width B	-	Light = 16 ftL	Width	---	0.65	mm
5223	Modulation	-	Light = 16 ftL	ΔEC	---	38	Vdc

NOTES:

1. This value is recommended only for low-velocity deflecting and low-ambient light levels.
2. All light dependent tests to be measured with a 1-1/8-inch (28.58 mm) vertical and 1-9/16-inch (39.69 mm) horizontal, 30-line raster. Photocell is to have a 1-inch (25.40 mm) active aperture and calibrated in foot lamberts similar to Photovolt Corporation type 200-A, or equal.
3. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 17, 80
DSA - ES

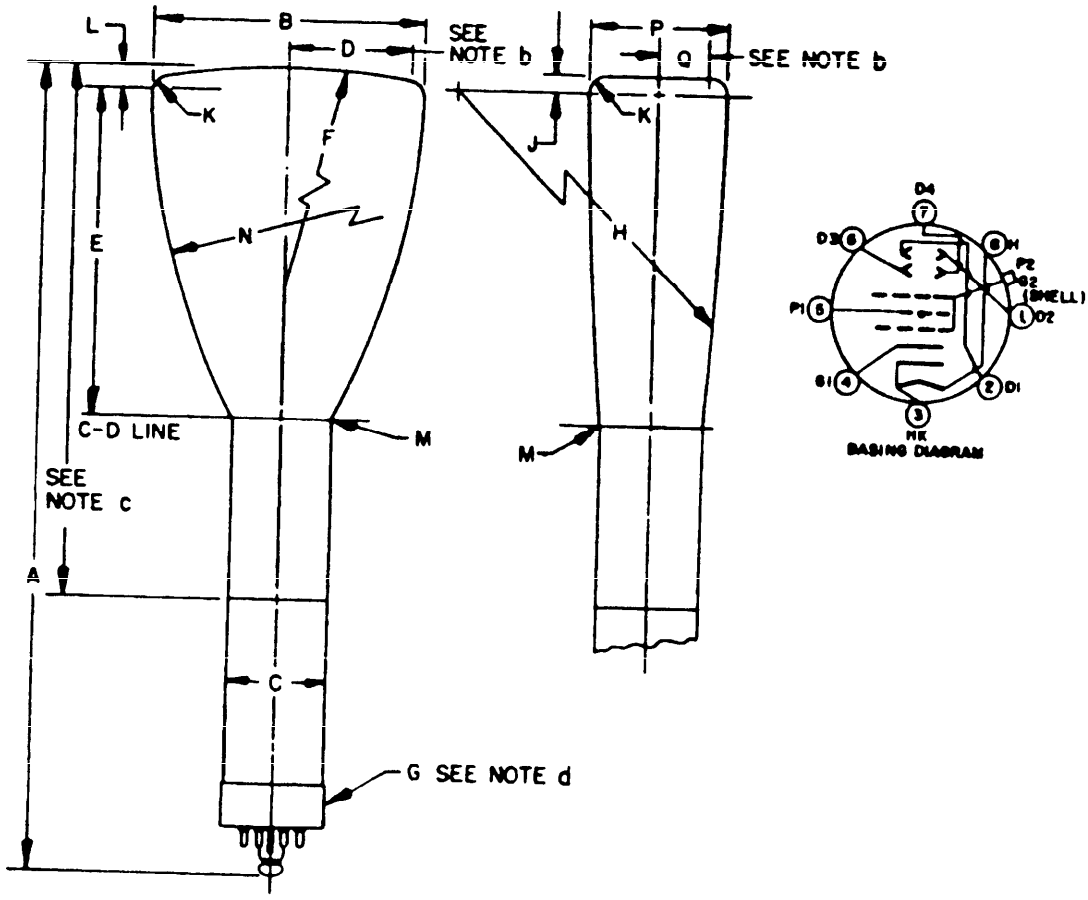
User activities:
Army - MC
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Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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NOTES:

a. These dimensions are for information only and are not required for inspection purposes.
b. Useful screen.
c. This portion of the envelope shall be coated with clear-baking synthetic XS-1810, as supplied by Coating Materials Laboratories, Inc., or equal.
d. Base: D8-1 (EIA). (Except the "D" dimension of the base shall be modified to "1.170(min) - 1.250(max)")

FIGURE 1. Outline drawing of electron tube type 3XP1.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
B	2.953 (75.01)	3.047 (77.39)
G	Base: D8-1 (EIA) (see note d)	
P		1.516 (38.51)
Quality conformance inspection, part 2		
A	8.750 (222.25)	9.000 (228.60)
C		1.125 DIA (28.57)
D	1.375 (34.93)	
Q	.563 (14.30)	
Reference dimensions (see note a)		
E	3.879 (98.53)	
F	15.000 R (381.00)	
H	35.500 R (901.70)	
J	.188 (4.78)	
K	.188 R (4.78)	
L	.246 (6.25)	
M	1.000 R (25.40)	
N	10.000 R (254.00)	

FIGURE 1. Outline drawing of electron tube type 3XP1 - Continued.

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